Muddana Ahalya

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Professional Summary

Recent B.Tech graduate in Information Technology with hands-on experience in **Python, SQL, machine learning, software development, and REST APIs**. Strong foundation in **OOP, data structures, DBMS, and backend development**, with experience using **Node.js, Express.js, Scikit-learn, TensorFlow, Git, and AWS**. Seeking an entry-level Python Developer role to apply programming and problem-solving skills to develop reliable software solutions.

Work Experience

AI-ML Virtual Intern Dec 2024 – Feb 2025

- Developed and evaluated machine learning and deep learning workflows using Python, covering data preprocessing, feature engineering, model training, testing, and performance evaluation.
- Built end-to-end machine learning pipelines involving data preparation, feature engineering, model development, testing, and model performance evaluation.
- Worked with AWS services including Amazon SageMaker, Amazon Bedrock, and Amazon Forecast for machine learning model development, deployment, and scalable AI solutions.

Projects

Smart Soil Nutrient and Moisture Monitoring System *Jan 2026 – May 2026*

IoT, Arduino, Modbus RS485, Node.js, Express.js, Python, Scikit-Learn, Chart.js

- Developed an IoT-driven monitoring system integrating Modbus RS485 NPK sensors and capacitive moisture sensors with Arduino to capture real-time soil nutrient telemetry.
- Integrated a Node.js and Express backend with a Python Scikit-Learn Decision Tree model to process live serial data and generate real-time, hourly, and daily crop suitability recommendations.
- Designed a responsive web dashboard using Chart.js to visualize real-time soil health metrics and enable automated CSV dataset logging and export for precision farming.

VecTorium: Hybrid Deep Learning Ensemble for Brain Tumor Classification *Oct 2025 – Nov 2025*

Python, TensorFlow, Keras, EfficientNetB3, ResNet50, VGG16, OpenCV, Scikit-Learn, NumPy, Matplotlib

- Engineered a custom deep learning ensemble model (VecTorium) integrating frozen pre-trained CNN backbones (EfficientNetB3, ResNet50, VGG16) with GlobalAveragePooling2D and feature concatenation for 4-class MRI brain tumor classification.
- Implemented real-time image preprocessing, data augmentation (*ImageDataGenerator*), dynamic learning rate decay (*ReduceLROnPlateau*), and *EarlyStopping* callbacks to optimize training and prevent overfitting.
- Evaluated model performance using classification metrics, training/validation loss curves, and confusion matrices, achieving 96.73% validation accuracy and 99.49% training accuracy.

Education

QIS College of Engineering and Technology, Ongole 2022 - 2026
B.Tech in Information Technology CGPA: 8.3/10

Sri Chaitanya Junior College, Addanki 2020 - 2022
Intermediate Education(MPC) 85.7%

ZPHS High School, Kondamanjuluru 2019 - 2020
Secondary Education(SSC) 100%

Technical Skills

Programming	Python, SQL, Java, HTML, CSS, JavaScript
Core Concepts	Data Structures, Object-Oriented Programming (OOP), DBMS, Computer Networks, Operating Systems, Software Development
Web Technologies	Node.js, Express.js, REST APIs, Responsive Web Development
Libraries	Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch
Tools & Platforms	Git, GitHub, Jupyter Notebook, VS Code, Arduino UNO, AWS

Certifications

- **Spring and Angular Full Stack Developer (Infosys)** **June 11, 2025**
- **Python Full Stack Developer (Eduskills)** **October – December 2024**
- **AWS Academy Cloud Foundations (Amazon)** **April 2025**
- **Python Programming (Codetantra)** **April 2025**

Achievements

- **Meri Maati Mera Desh Programme**